

CSA 1718 – Artificial Intelligence

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Aim

To write and execute a Prolog program to find the sum of integers from **1 to N** using recursion.

Objective

- To understand recursion in Prolog.
- To calculate the sum of natural numbers from 1 to a given number N.
- To understand facts, rules, and queries in Prolog.

Theory

In Prolog, recursive rules can be used to calculate the sum of integers from 1 to N.

The mathematical formula is:

$$\text{Sum} = 1 + 2 + 3 + \dots + N$$

The recursive definition is:

- $\text{sum}(0, 0) \rightarrow$ Base case
- $\text{sum}(N, S) \rightarrow$ Sum of numbers from 1 to N

For example:

$$\text{Sum of 1 to 5} = 1 + 2 + 3 + 4 + 5 = 15$$

Prolog Program

% Sum of integers from 1 to N

$\text{sum}(0, 0).$

$\text{sum}(N, S) :-$

$N > 0,$

$N1 \text{ is } N - 1,$

$\text{sum}(N1, S1),$

S is $S1 + N$.

Explanation of the Program

1. Base Case

sum(0, 0).

When $N = 0$, the sum is 0.

2. Recursive Rule

sum(N, S) :-

$N > 0$,

$N1$ is $N - 1$,

sum($N1$, $S1$),

S is $S1 + N$.

- $N > 0$ checks that the number is positive.
- $N1$ is $N - 1$ decreases N by 1.
- sum($N1$, $S1$) recursively calculates the sum up to $N-1$.
- S is $S1 + N$ adds the current value N to the previous sum.

Sample Query

?- sum(5, S).

Output

S = 15.

Another example:

?- sum(10, S).

Output:

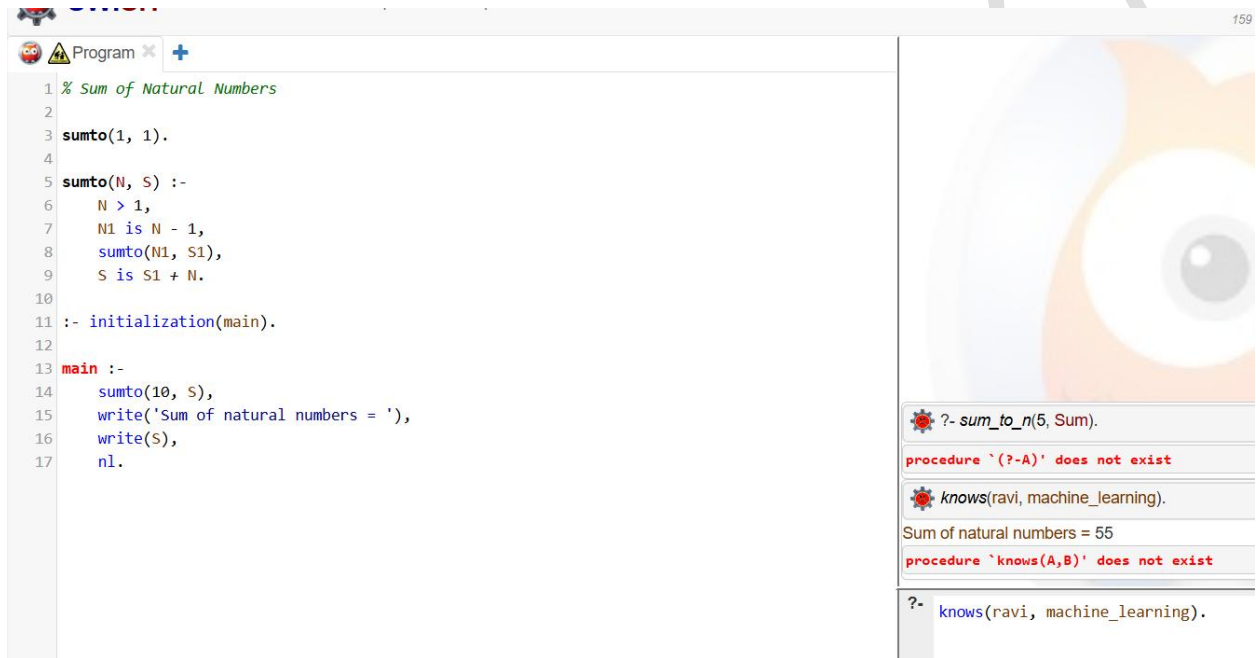
S = 55.

Algorithm

1. Start.
2. Read the value of N .
3. If $N = 0$, return 0.

4. Otherwise, calculate $N - 1$.
5. Recursively find the sum from 1 to $N-1$.
6. Add N to the obtained sum.
7. Display the final sum.
8. Stop.

Out Put



```
1 % Sum of Natural Numbers
2
3 sumto(1, 1).
4
5 sumto(N, S) :-
6     N > 1,
7     N1 is N - 1,
8     sumto(N1, S1),
9     S is S1 + N.
10
11 :- initialization(main).
12
13 main :-
14     sumto(10, S),
15     write('Sum of natural numbers = '),
16     write(S),
17     nl.
```

?- sum_to_n(5, Sum).
procedure `(?-A)' does not exist
?- knows(ravi, machine_learning).
Sum of natural numbers = 55
procedure `knows(A,B)' does not exist
?- knows(ravi, machine_learning).

Result

The Prolog program was successfully executed to calculate the **sum of integers from 1 to N using recursion.**